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Intro Inverse Problems and Imaging (02624)

Week 3

Hilbert spaces and compact operators

Hilbert spaces X : Complete space with inner product $(x_1, x_2) \in \mathbb{R}$

Examples

- Euclidean space $X = \mathbb{R}^n$ with inner product $(x_1, x_2) = x_1 \cdot x_2 = x_1^T x_2$
- Function space $X = L^2(a, b)$ with inner product

$$(x_1, x_2) = \int_a^b x_1(s) x_2(s) \, \mathrm{d}s.$$

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Compact operator $K \in B(X, Y)$ from Hilbert spaces X to Y. Think of generalization of matrices to infinite dimensions.

Examples:

$$A \in \mathbb{R}^{n\times m}, \quad Kx(t) = \int_a^b k(t, s)x(s) \, \mathrm{d}s \text{ in } L^2(a, b) \text{ with } \int\int |k(s, t)|^2 \mathrm{d}t\mathrm{d}s < \infty.$$

Singular Value Decomposition

X, Y : Hilbert spaces. Compact operator $K : X \to Y$.

Assume $\text{Dim}(X) = \infty$, $\mathrm{Ker}(K) = \{0\}$. Singular Value Decomposition

$$\{x_j\} \text{ ONB for } X, \qquad \{y_j\} \text{ ONB for } \overline{\text{Ran}(K)} = \text{Ker}(K^*)^\perp$$
$$Kx_j = \mu_j y_j$$

with

$$0 \leq \mu_{j+1} \leq \mu_j \text{ with } \lim_{j\to\infty} \mu_j = 0.$$

Expansion

$$Kx = \sum_j \mu_j (x, x_j) y_j, \qquad \text{for } x = \sum_j (x, x_j) x_j.$$

Example: Numerical differentiation in $L^2(0,1)$

$$Kx(t) = \int_0^t x(s)\,\mathrm{d}s$$

Singular system

$$x_j(s) = \sqrt{2}\cos\left(\left(j+\frac{1}{2}\right)\pi s\right)$$

$$y_j(t) = \sqrt{2}\sin\left(\left(j+\frac{1}{2}\right)\pi t\right)$$

$$\mu_j = \frac{1}{\left(j+\frac{1}{2}\right)\pi}, \qquad j = 1, 2, \ldots$$

The role of noise

Data is always corrupted by noise

$$y^\delta = Kx + n = y + n$$

Absolute noise level $\|n\| = \delta$ Relative noise level $\frac{\|n\|}{\|y\|} = \frac{\delta}{\|y\|}$.

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Two problems may arise:

1. $\mathrm{Proj}_{\mathrm{Ker}K^*}(e) \neq 0$
2. $e - \mathrm{Proj}_{\mathrm{Ker}K^*}(e) \notin \mathrm{Ran}(K)$

NB:

- In infinte dimensions, $\mathrm{Ran}(K)$ is not closed.
- Assume from now $\mathrm{Ker}K^* = \{0\}$.

The Picard condition

For

$$x = \sum_j (x, x_j) x_j$$

consider

$$Kx = y$$

$$Kx = \sum_j (x, x_j) K x_j = \sum_j \mu_j (x, x_j) y_j = \sum_j (y, y_j) y_j = y.$$

Solvable if and only if

$$y \in \overline{\mathrm{Ran}\, K} = (\ker K^*)^{\perp} \quad \text{and} \quad \sum_j \frac{|(y, y_j)|^2}{\mu_j^2} < \infty.$$

The Picard plot!

Truncated SVD - TSVD

Solving

$$Kx = \sum_j \mu_j (x, x_j) y_j = \sum_j (y, y_j) y_j = y$$

gives

$$(y, y_j) = \mu_j (x, x_j) \quad \text{(for all } j\text{);} \qquad x = \sum_j \frac{(y, y_j)}{\mu_j} x_j$$

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Division by $\mu_j \ll 1$ is problematic

Truncated SVD with $\alpha > 0$

$$x_\alpha = R_\alpha y = \sum_{\mu_j^2 \geq \alpha} \frac{(y, y_j)}{\mu_j} x_j.$$

Matrix SVD

For $A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^n, y \in \mathbb{R}^m$:

$$A = USV^T$$

with orthogonal matrices (column vectors v_j, u_i)

$$V = [v_1 v_2 \cdots v_n] \in \mathbb{R}^{n \times n} \quad U = [u_1 u_2 \cdots u_m] \in \mathbb{R}^{m \times m}$$

and “diagonal matrix”

$$S = \mathrm{diag}(\mu_j) \in \mathbb{R}^{n \times m}.$$

Truncated SVD with $\mu_k^2 \geq \alpha > \mu_{k+1}^2$

$$x_\alpha = \sum_j^k \frac{u_j^T y}{\mu_j} v_j = V(:, 1:k) S(1:k, 1:k)^{-1} U(:, 1:k)^T y.$$